Exploring a Use-case for AI-based Services in Banking: Project Regulatory Compliance by Design

PGDip Applied Data Science

RESAC9013 Research Methods with Ethics and Data Governance

Continuous Assessment (CA3)

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Abstract

The topic of this research is *Exploring a Use-case for AI-based Services in Banking: Project Regulatory Compliance by Design.*

This paper primarily builds on a design concept from a specific research (1) on the identification of legal and regulatory obligations for banking compliance, and their specific relevance to individual banking procedures, and considers an end-to-end logical design of a proposed solution. In terms of the banking sector, banks have a significant challenge to ensure and evidence that they comply with legal and regulatory obligations. Both external obligations and internal procedures change at pace and hence continuous validation is of material importance. The consequences of failure are considered a critical risk with materially important consequences. The proposed solution design is a logical end to end consideration of identifying and sourcing valid documents, encoding document contents based on Large Language Model algorithm, and determining the relevance of specific obligations to specific banking procedures. It is noted that there has been no attempt to explore existing solutions to this problem available on the market, hence this is an area of important further research.

1 Introduction

The rules by which Irish banks are governed are set by a combination of EU legislation and national laws.(2). As Ireland is an EU Member State, the competent authority for banking regulation is the European Central Bank, which directly supervises companies designated as 'significant institutions', with the Central Bank of Ireland supervising 'less significant institutions', with regulation being mainly defined by the Single Supervisory Mechanism Mechanism (SSM) Framework Regulation and the Central Bank Acts. Supervision controls that a bank complies with its legal and regulatory obligations, with the sanctions for the company, and its personnel, for non-compliance.

At the start of 2025 there were 442 credit institutions listed on the Central Bank of Ireland's register of credit institutions, of which 18 were Irish banks and 30 were EEA authorised banks with a branch in Ireland (3).

The legal and regulatory environment is not static, and banks must not only comply with current obligations, but also scan the horizon for upcoming changes and new obligations.

A bank must ensure its day-to-day operations comply with its obligations. Additionally, When a bank updates its products, services or ways of working it must ensure that all changes are also in line with its obligations.

With respect to legal and regulatory compliance, a key challenge for banks is to identify all of its obligations, and to understand which obligations relate to which of its internal

procedures. This is typically a time-consuming, manual activity performed by experts in specific banking functions such as Risk and Compliance departments.

It is of vital importance to have accurate interpretation of relevant obligations for each operational activity, in order to ensure compliance.

The aim of this report is to understand how an application of Artificial Intelligence can assist in the identification and interpretation of banking obligations and the relevance of each in the context of internal operational activities.

The report considers the problem, the data available to a bank, and a proposed AI solution. Ethical considerations for the AI solution are considered, along with an appropriate data governance framework.

2 Problem Statement

2.1 Challenges

The identification and interpretation of current and up-stream legal and regulatory obligations is not a new challenge for banks, and is a vital assurance activity of a Risk & Compliance function.

There is no 'one-stop-shop' to easily source all relevant legal and regulatory documents with which a bank must comply. People with the right skills and experience must keep up to date as obligations evolve and change over time. Each must be interpreted and considered in the context of all of the bank's products, services and operational activities to ensure everything that the bank does is in compliance with the rules within which it is obliged to conduct itself.

A bank formalises how it conducts its business by defining an extensive set of policy documents approved by its Board of Directors. These are a bank's internal rule-book. Each policy is further detailed into specific processes and detailed procedural tasks that explicitly demonstrate how its operational activities are performed. The processes and procedures have to be tested to ensure they comply with all relevant obligations.

2.2 Requirements

Hence, there are five main requirements to be considered as part of the solution to the problem:

1. compiling and maintaining a complete list of all legal and regulatory obligations;
2. compiling and maintaining a complete list of all internal policies and procedures;

3. matching which specific legal and regulatory obligations are relevant to each internal process and its underlying procedural tasks;
4. being able to search for relevant obligations for a specific aspect of its operations (useful when a process is being reviewed/changed);
5. being able to search for relevant policies, processes and procedural tasks for a specific legal or regulatory obligation (useful when a law or regulation changes, to know what aspects of operations will need to be reviewed).

It is also important to note that there are a variety of commercial tools available to banks to assist them in tracking their obligations. The study for this paper has not included a research of what tools and features are available in the marketplace. This is considered an important area of further research prior to progressing the implementation of the solution proposed in this paper.

3 Proposed AI-Based Solution

3.1 Solution Design - Level 0 Context

At a context level of design, the overall solution can be summarised with two inputs and two main processes (refer Figure 1):

1. sourcing the latest legal and regulatory documents that make-up the bank's obligations;
2. sourcing the latest internal procedural document of the banks, defining the way the bank operates;
3. determining the relevance of each obligation with each of the bank's procedures;
4. based on an input search parameter (obligation or procedure), show relevant corresponding obligations or procedures.

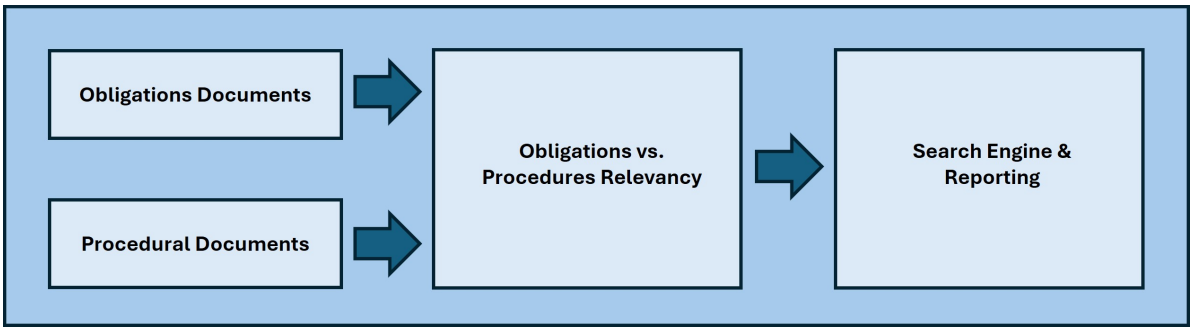

Figure 1: Solution Design - Level 0 Context

3.2 Solution Design - Level 1 & Level 2 Detail

It is helpful to break down the solution design into the next level of detail (refer Figure 2).

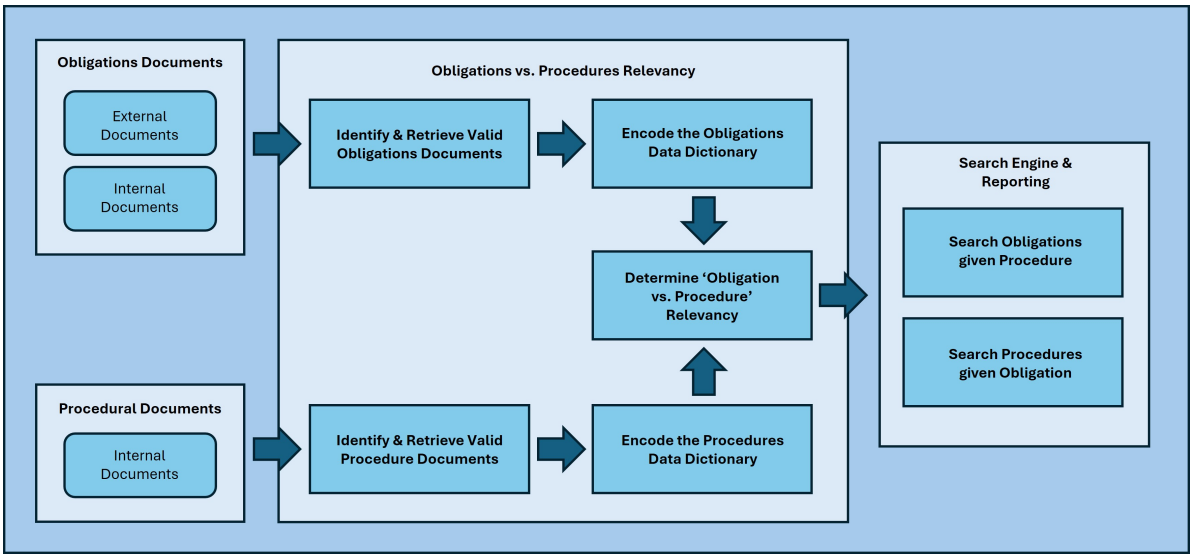

Figure 2: Solution Design - Level 1 Detail

3.2.1 Obligations Documents

The obligations are sub-categorised into external obligations and internal obligations.

External Obligations

External obligations are comprised of a number of legal and regulatory regulations and directives.

They can be sourced on the internet from a variety of official online sites, for example:

- European Banking Authority
- European Commission
- European Parliament
- Central Bank of Ireland

Each location will include a link to one or more obligation 'paper', often, but not always, encoded in Portable Definition Format (PDF). Each paper may contain relevant obligations by which the bank must comply.

Internal Obligations

Internal obligations are defined by the bank themselves, and take the form of Board approved Policy Documents. A policy is an internal 'rule' often transposed from an external obligation, but potentially augmenting external obligations or clarifying explicitly how an external obligation must be fulfilled.

The internal obligations are necessary as there is a level of interpretation of an external obligation. Hence by confirming the bank's interpretation through defining the internal obligation removes ambiguity as it confirms the interpretation of the bank.

3.2.2 Procedural Documents

The bank defines internal documents outlining what each organisational unit (function) is responsible for, and the processes they own (and contribute to).

Each process is described and broken down to the step by step procedure that must be followed. In this way activities are standardised and expected to be performed in a prescribed way. Each process formalises where controls are performed to mitigate risk.

The lowest level of a procedure is an individual task.

Standard ways of codifying the primary activities of a bank through functional processes and procedures ensure that there is an internal 'playbook' of how the bank operates. These internal documents describe the bank's procedures.

When a bank formulates its processes and procedures, it has to take into account the relevant obligations, and ensure the procedure complies with those obligations.

3.2.3 Obligations vs. Procedures Relevancy

This design component can be broken down into five stages:

- *Identify & Retrieve Valid Obligations Documents:* The solution must interrogate a potentially valid document source location, identify potentially valid documents, retrieve those documents to a staging location, test those documents to determine if they have valid obligations vs. a set of defined validity criteria. If the document is considered to be a valid source of 'obligations' then its contents will be encoded in the Obligations Data Dictionary (next step).
- *Encode the Obligations Data Dictionary:* For each valid document sourced, it will be decomposed into a set of 'chunks', for example, sentences. Each of these will be converted to a mathematical vector, and will be tested against the existing contents of the Obligations Data Dictionary to determine the action to take, i.e. create a new entry in the database, update an existing entry in the database, or discard the chunk

(no update to the database required, for example it may already exist). Refer Figure 3).

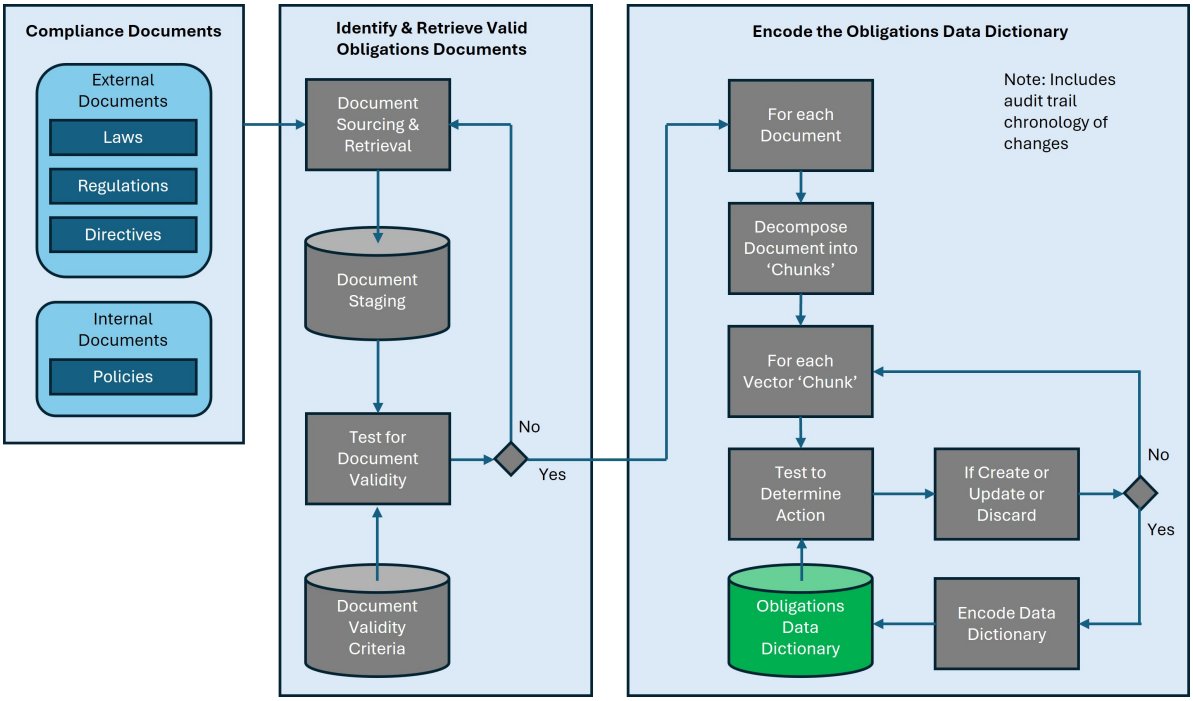

Figure 3: Solution Design - Source & Encode Obligations - Level 2 Detail

- *Identify & Retrieve Valid Procedure Documents:* In a similar way as for Obligations Documents, the solution must identify potentially valid internal procedural documents, retrieve those documents to a staging location, and test those documents to determine if they have materially important processes and procedures (based on a set of defined validity criteria). If the document is considered to be a valid source of 'procedures' then its contents will be encoded in the Procedure Data Dictionary (next step).

- *Encode the Procedures Data Dictionary:* For each valid procedure document sourced, it will be decomposed into a set of 'chunks', for example, procedural tasks. Each of these will be converted to a mathematical vector, and will be tested against the existing contents of the Procedure Data Dictionary to determine the action to take, i.e. create a new entry in the database, update an existing entry in the database, or discard the chunk (no update to the database required, for example it may already exist). Refer Figure 4).

- *Determine 'Obligation vs. Procedure' Relevancy:* The Obligations Data Dictionary, and the Procedures Data Dictionary, are both maintained on a suitably regular frequency. This design component is responsible for matching a procedural entry from the Procedure Data Dictionary, and to obligation entries (from the Obligations Data Dictionary). The matching is not binary matched/non-matched. As noted earlier, there

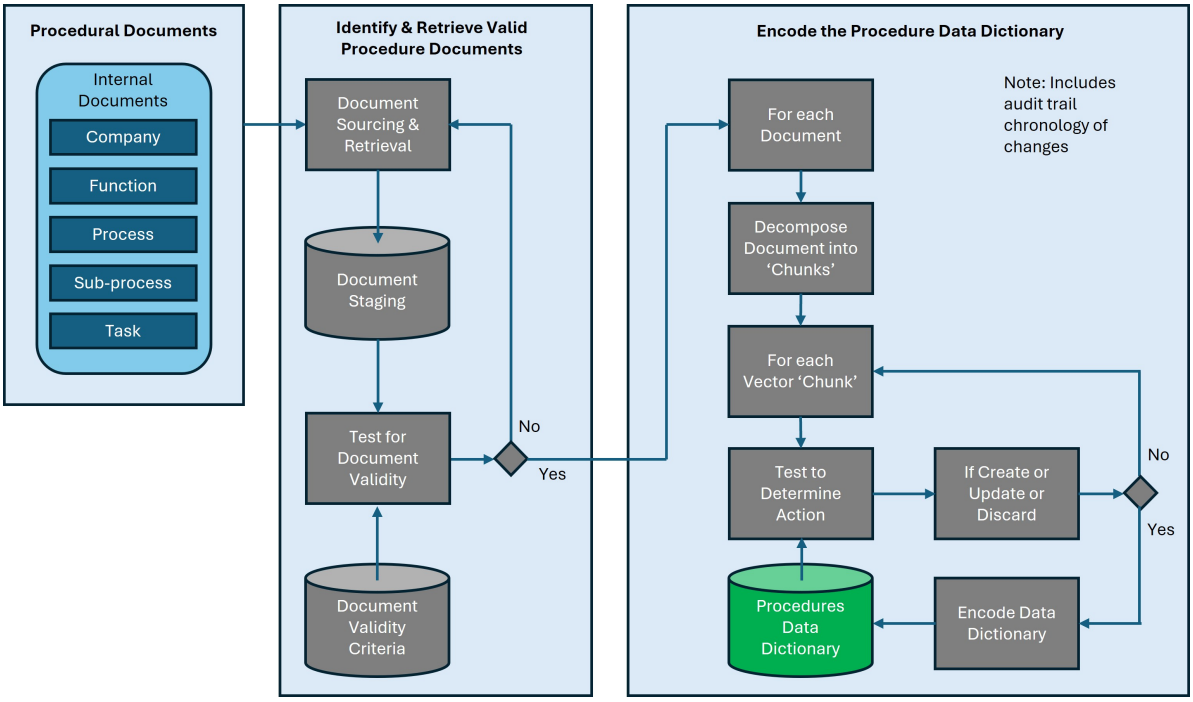

Figure 4: Solution Design - Source & Encode Procedures - Level 2 Detail

is a level of 'interpretation' of an obligation in the way it is transposed into a bank's operations. Hence, a 'score' is calculated to show the level of relevancy of each obligation to a specific procedure. The relevancy score for a procedure entry and obligation entry pair is stored as a mathematical vector in the 'Relevancy Data Dictionary'. Refer Figure 5).

Relevancy can be calculated for obligations matched to processes, sub-processes, procedure, and down to the lowest level of tasks.

3.2.4 Search Engine & Reporting

All of the previous design components relate to sourcing and encoding information in a database. The value of doing this is realised through this design component which enables a) search and b) reporting functionality. Refer Figure 6).

Search Engine

A user can enter parameters into a digital input form, which will be used to search for a specific obligation(s) and identify matching procedures (with relevancy), or to search for a specific procedure(s) and identify matching obligations (with relevancy).

An example use case for 'search' is that a Business Analyst may be in the process of reviewing

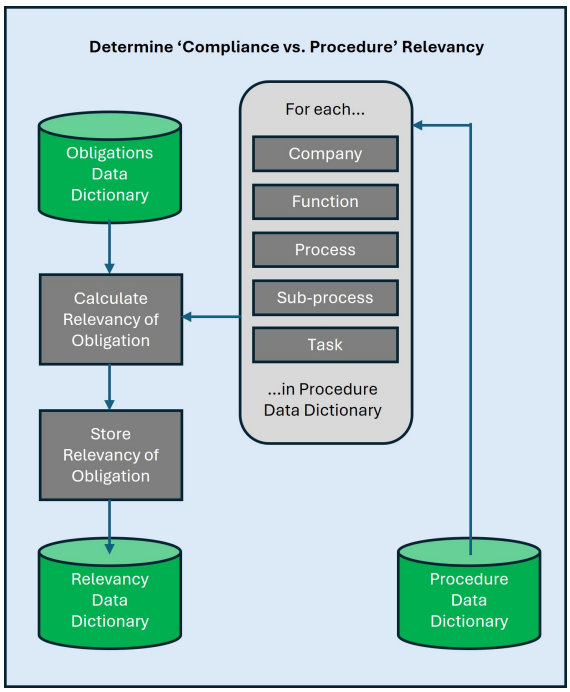

Figure 5: Solution Design - Determine 'Obligation vs. Procedure Relevancy' - Level 2 Detail

a specific internal procedure, where it is invaluable for them to be able to check which are the latest version of all associated obligations, and test the procedural compliance as part of the operational workflow design.

Reporting

Reporting is similar to Searching except some of the query parameters may be standardised and the output report format is standardised in each case.

An example use case for 'reporting' is that the Internal Audit department (3rd line of defense) may be preparing for an audit of a particular function, and may want a complete view of all obligations related to that department, as well as procedural relevancy as a basis for the audit design.

3.3 Viable Datasets

A complete list of viable source locations to identify potentially valid obligations documents has not been performed, although a number of illustrative examples are provided below:

1. European Banking Authority site - Interactive Single Rulebook (4)
2. European Commission site - European Artificial Intelligence Act 2024 (5)

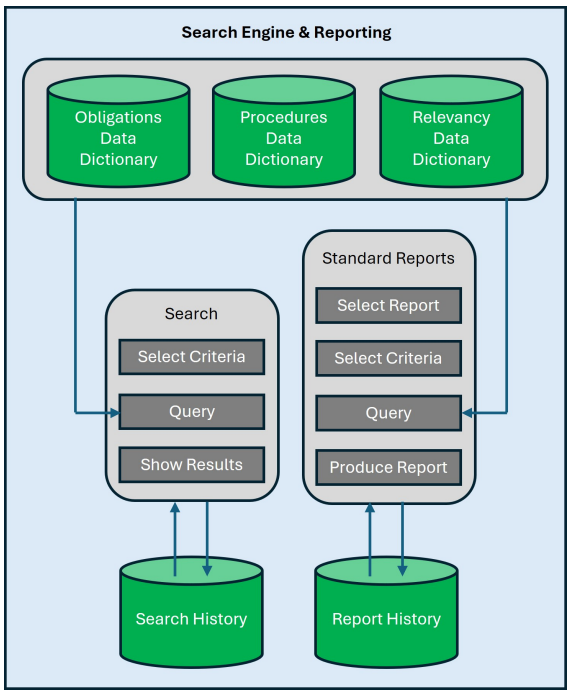

Figure 6: Solution Design - Search Engine & Reporting - Level 2 Detail

3. Irish Government site - General Data Protection Act 2028 (6)
4. Central Bank of Ireland site - Regulatory Requirements and Guidance for Credit Institutions (7)

The actual number of obligation documents is expected to be large (well in excess of 100) and hence the number of individual 'chunks' for encoding in the obligations register could be huge (assumption could be 500 documents averaging 50 pages each, with potential for average of 1000 obligation 'chunks' per document i.e. order of magnitude 500,000 individual obligations.

3.3.1 Dataset Tables

Obligations Data Dictionary

An example of the metadata attributes for an obligation entry in the Obligations Data Dictionary could be defined as follows:

- Obligation ID (Integer)
- General Document Name (String)
- Specific Document Name (String)
- Document Date (DateTime)

- Authorising Entity (Categorical String)
- External Source Location URL (String)
- Sourced Date (DateTime)
- Internal Staging Location URL (String)
- Obligation Vector Description (Tuple)
- Obligation Vector Value (Tuple)

Procedures Data Dictionary

An example of the metadata attributes for a procedure entry in the Procedures Data Dictionary could be defined as follows:

- Process ID (Integer)
- Sub-process ID (Integer)
- Procedure ID (Integer)
- Task ID (Integer)
- Process Owner E-mail (String)
- Sub-process Owner E-mail (String)
- Procedure Owner E-mail (String)
- Task Owner E-mail (String)
- Department ID (Integer)
- Function ID (Integer)
- Team ID (Integer)
- Specific Document Name (String)
- Document Date (Datetime)
- Document Status (Categorical String)
- Internal Source Location URL (String)
- Sourced DateTime (DateTime)
- Internal Staging Location URL (String)
- Procedure Vector Description (Tuple)
- Procedure Vector Value (Tuple)

Relevancy Data Dictionary

An example of the metadata attributes for an entry in the Relevancy Data Dictionary could be defined as follows:

- Process ID (Integer)
- Sub-process ID (Integer)
- Procedure ID (Integer)
- Task ID (Integer)
- Obligation ID (Integer)
- Relevancy Score (Float)
- Relevancy Vector Description (Tuple)
- Relevancy Vector Value (Tuple)

4 Ethical Considerations

Based on a previous paper as part of this course of study (8), a large number of potential ethical concerns were identified, categorised thematically under a) Unfair Customer Outcomes, b) Data Processing & Management, and c) Legal & Regulatory Obligations.

Consideration 01: Automated Relevancy Scoring

A key design concept of the solution is the automated sourcing of documents, document validity determination, and encoding of obligations and procedures, as well as the automated relevancy scoring of matching obligations to procedures.

We have previously noted that obligations have to be interpreted as to how they will be enacted into operations. The method of 'interpretation' is a key component of assessing the 'relevancy' of an obligation to a procedure. This is not an exact science (matching an explicit value with another explicit value) and hence requires materially important model training and evaluation.

The primary research paper on this concept (1) concluded from its results that there are significant challenges in transparency, reproducability and accuracy of results. This is a key concern and would require considerable uplift in quality of results. Accuracy is very important, if the solution is intended to replace the largely manual effort of Compliance experts, and incorrect results could mean banks consider their procedures to be compliant, when in fact they are not. This has material risk to regaulatory sanction and reputational damage.

Consideration 02: Unethical Business Behaviour

It could be considered an 'excuse' for non-compliance if the solution does not indicate a *relevancy*, when in effect there should be one. On this basis, there is considered a high importance for human oversight of the results of the model.

Consideration 03: Bias in Algorithms

The quality of obligations and procedures encoding, and relevancy scoring, is only as good as the data on which the model is trained. There could potentially be bias in the underlying data based on the legal jurisdiction (US, Europe, other), as certain obligations are specific to certain jurisdictions.

Similarly, there could be bias in the specification of the validity criteria and the relevancy scoring.

Consideration 04: Equity and Fairness

Whilst it has been proposed to determine a *relevancy score* to match an obligation with a procedure, it has not been proposed to determine an *importance score* for each obligation. In reality, some obligations may be more materially important and consequential than others. It is therefore considered least risky to not attempt to qualify the importance of an obligation.

On this basis we make no attempt to evaluate the potential consequence of non-compliance of an obligation, and this is potentially an area of further proposed research.

Consideration 05: Data Traceability

The solution has proposed that every document source is stored as metadata, as well as storing the physical document in a staging area, and not relying on the online version.

There is no assurance that documents sourced at points in time will remain available online. If a more recent version of a document is published, the previous version may no longer be accessible online.

It is considered ethically important that if any evaluation is done in terms of obligations vs. procedure relevancy, then the physical version of teh source documents on which the evaluation was made should always be available for audit purposes, up to the threshold for document retention as defined in bank policies.

Consideration 06: Data Limitations

The solution relies on sourcing *all* obligation documents applicable to a bank. If this is incomplete, then there is potential for non-compliance (unknown-unknowns).

Consideration 07: Data Accuracy

There is an assumption that the documents sourced are reliable in terms of data accuracy. This is of the utmost importance. The source obligation documents will need to be validated by a human expert to ensure they are a reliable source.

Similarly, the accuracy of the encoding of internal procedures needs to be ensured. The application of standardised methods for documenting internal procedures, and quality control, is an important consideration to ensure accuracy.

Consideration 08: AI Implementation & Governance Frameworks

The design, build and test methodology of the solution must be fully integrated with the bank's AI Implementation & Governance framework to ensure its ethical design and adoption, and assurance of adequate testing.

Consideration 09: Data Privacy

The external obligations source documents are all publicly available. However, the internal obligations source documents (policies) and the internal procedures documents (the bank's own operational processes and procedures) are confidential and the data should not be exposed to publicly available large language models.

There is potential for personal data to be exposed, e.g. name of process owner, name of policy owner, and their e-mail addresses, although this could be anonymised by storing generic data such as person's role, as opposed to person's name or e-mail address.

5 Data Governance Framework

Data Governance is a material design component for this solution. There are a number of considerations:

Data Creation/Update/Deletion

There must be a defined policy for the creation, update and deletion of data at the key stages of the solution workflow:

- Document Sourcing
- Document Staging
- Document Evaluation
- Document Metadata

- Document Decomposition

LLM Algorithm Training & Testing

The embedded algorithms (document sourcing, document validity, document encoding, obligation vs procedure relevancy evaluation) must be trained on a a sample dataset of relevant external and internal documents, where the data has been properly reviewed and cleansed to ensure accuracy. Data with quality issues should be introduced in a controlled way,to train the model to identify potential quality issues.

Data Security & Privacy

The publicly available sourced data is not a concern.

The internally sourced data (policies and procedures) as well as processed data (relevancy scores, search results, reports) must be considered commercially sensitive and private and hence be managed under strict policy guidelines including all legal and regulatory obligations.

Data Retention

No data should be deleted from the system unless it was stored erroneously. Historic data can be archived, for performance reasons, but should not be deleted, as historic information and comparisons may be part of reporting functionality e.g. plot the increase in number of credit related obligations over the last 'n' years.

Data Availability & Access Control

The solution should be available to any employee to use search functionality. Access to specific standard reports can be on a case by case basis (report owner approves access request).

Model Performance

The obligations encoding component, and the procedures encoding component can be run as background processes, and the frequency by which the solution should search for and source valid documents is an important parameter to be defined for example, once per week. These background processes could be very time-consuming and consume much processor memory.

Similarly, the matching of obligations vs. procedures and relevancy evaluation is a background process that should be performed whenever a new document has been sourced.

Foreground event driven processes are search and reporting functions. These need to be fast, efficient, and accurate, and hence processor must prioritise these functions over background

processing.

6 Conclusion

This paper has considered existing research on the identification of obligations relevant to banking procedures, and has considered an end-to-end logical design of the solution.(1)

In terms of the banking sector, banks have a significant challenge to ensure and evidence that they comply with legal and regulatory obligations. The consequences of failure are a critical risk, ultimately senior executives can face criminal proceedings for failure, and a bank can have very high financial penalty penalties (fines) imposed, and can have the banking license removed.

The process of ensuring compliance requires experts and is time-consuming, and a manual effort. It is not fail-safe and errors occur.

The obligations are not static. They evolve over time, and in general teh volume of obligations requiring compliance increases with time.

The internal procedures are not static. Banking operations continuously improve and the internal procedures evolve over time.

Staff turnover implies the importance that internal procedures are properly documented and updated to reflect changes. There can be no reliance on the knowledge of individual staff members. The same goes for staff knowledge of obligations - they must be documented with no reliance on knowledge or experience of individuals.

Interpretation of obligations, and how internal procedures are considered to comply with each obligation, are subjective, and having a framework to encode this in a consistent form removes bias from the results, and improves quality.

There are systems on the market that assist with tracking obligations and determining business process relevancy, example 'corlytics' (9). This paper has not performed any review of the functionality of these systems compared with the proposed design of the solution in this paper, which has been done in isolation. This is certainly an area for further research.

Areas for future research include:

- Criteria to identify relevant obligation source sites
- Criteria to identify relevant obligation documents
- Materially important attributes for obligation and procedure vector definition
- Approach to determining relative importance (and consequence) of each obligation
- Improving reliability of specific solution algorithms through training

- Determining an inventory of existing commercial solutions for obligations tracking and business procedure relevancy, and comparison on effectiveness, and gaps to be addressed through further research

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